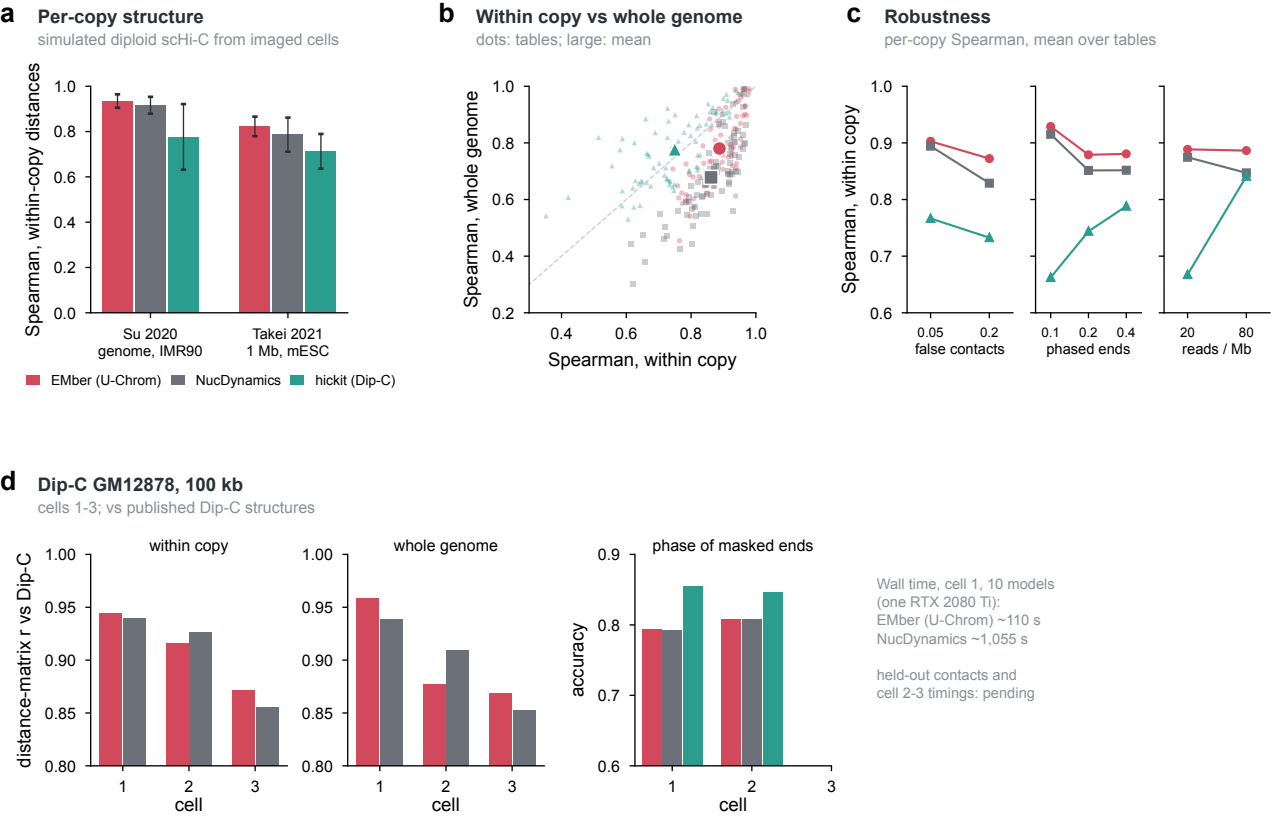

a
Per-copy structure
simulated diploid scHi-C from imaged cells
Spearman, within-copy distances
1.0
0.8
0.6
0.4
0.2
0.0
Su 2020
genome, IMR90
Takei 2021
1 Mb, mESC
EMber (U-Chrom)
NucDynamics
hickit (Dip-C)
b
Within copy vs whole genome
dots: tables; large: mean
Spearman, whole genome
Spearman, within copy
1.0
0.8
0.6
0.4
0.2
0.4
0.6
0.8
1.0
c
Robustness
per-copy Spearman, mean over tables
Spearman, within copy
1.0
0.9
0.8
0.7
0.6
0.05
0.2
false contacts
0.1
0.2
0.4
phased ends
20
80
reads / Mb
d
Dip-C GM12878, 100 kb
cells 1-3; vs published Dip-C structures
within copy
whole genome
phase of masked ends
distance-matrix r vs Dip-C
accuracy
1.00
0.95
0.90
0.85
0.80
0.9
0.8
0.7
0.6
1
2
3
cell
Wall time, cell 1, 10 models
(one RTX 2080 Ti):
EMber (U-Chrom) ~110 s
NucDynamics ~1,055 s
held-out contacts and
cell 2-3 timings: pending